

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 4 : Session – 7 : SLO – 1 | Multiprocessing Using IPython

■ I. Fill in the Blanks

1. The multiprocessing module — it spawns separate OS processes so Python code runs truly in parallel (bypassing the GIL).
2. Pool — a Pool manages a group of worker processes and distributes tasks among them.

■ II. Multiple Choice

1. [1, 4, 9] — `p.map(square, [1, 2, 3])` applies `square()` to each element across 2 worker processes and returns the results in order.

■ III. Key Notes

- `Pool(n)` creates `n` worker processes; the `with` block ensures workers are closed and joined.
- `map()` preserves input order regardless of which process finished first.